Derived Performance Metrics Design

Overview

The objective of these derived metrics is to transform raw Fruit Ninja gameplay data into meaningful indicators of cognitive performance. Each metric is normalized to a scale of **0–100**, where a higher score generally indicates better performance (except Error Rate, where a higher value indicates more mistakes).

The formulas are designed to measure five key behavioral dimensions:

1. Accuracy
2. Response Speed
3. Error Frequency
4. Persistence
5. Consistency

An Overall Performance Score combines these dimensions into a single interpretable value.

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1. Accuracy Rate

Formula

Accuracy Rate = 100 × Fruits Sliced / (Fruits Sliced + Fruits Missed + Bombs Hit)

Why this Formula?

Accuracy measures how effectively the player interacts with valid targets while avoiding mistakes.

A player demonstrates good accuracy when they:

- Slice fruits successfully
- Miss fewer fruits
- Avoid hitting bombs

The denominator represents all meaningful slicing opportunities and mistakes, while the numerator represents successful actions.

Interpretation

- High fruits sliced → Higher accuracy
- High fruits missed → Lower accuracy
- High bombs hit → Lower accuracy

Example

Fruits Sliced = 90

Fruits Missed = 10

Bombs Hit = 5

Accuracy = $100 \times 90 / (90 + 10 + 5)$

Accuracy = 85.71

Cognitive Skill Measured

- Visual attention
- Hand-eye coordination
- Target recognition
- Precision

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2. Response Rate

Formula

Actions Per Second = (Fruits Sliced + Bombs Dodged) / Session Duration

Speed Score = min(100, 100 × Actions Per Second / 2.5)

Combo Score = min(100, 100 × Max Combo / 50)

Response Rate = 0.6 × Speed Score + 0.4 × Combo Score

Why this Formula?

Response Rate evaluates how quickly and effectively a player reacts to gameplay events.

Two important indicators are used:

Speed Score

Measures how many successful actions the player performs per second.

A faster player is usually responding more quickly to visual stimuli.

Combo Score

A high combo streak indicates the player can continuously react correctly without interruption.

Why 60% Speed and 40% Combo?

Reaction speed is the primary requirement of Fruit Ninja.

Combo performance is important but serves as supporting evidence of sustained reaction quality.

Interpretation

- Faster actions → Higher score
- Longer combo streaks → Higher score
- Slow gameplay → Lower score

Cognitive Skill Measured

- Reaction time
- Processing speed
- Visual-motor coordination

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3. Error Rate

Formula

Error Rate = min(100, 100 × (Fruits Missed + 3 × Bombs Hit) / (Fruits Sliced + Fruits Missed + Bombs Hit + Bombs Dodged))

Why this Formula?

Error Rate measures the frequency and severity of mistakes.

Two types of errors exist:

Fruits Missed

The player failed to react.

Bombs Hit

The player reacted incorrectly.

Since hitting a bomb is usually more severe than missing a fruit, bombs are weighted three times higher.

Bombs Hit Weight = 3

Interpretation

- More missed fruits → Higher error rate
- More bomb hits → Much higher error rate
- Fewer mistakes → Lower error rate

Cognitive Skill Measured

- Impulse control
- Decision making
- Selective attention

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4. Persistence Rate

Formula

Retries Score = 100 × Retries / 3

Duration Score = 100 × Session Duration / 300

Pause Penalty = 20 × Pause Count / 5

Persistence Rate = max(0, min(100, 0.5 × Retries Score + 0.5 × Duration Score - Pause Penalty))

Why this Formula?

Persistence measures willingness to continue engaging with a task despite challenges.

Three signals are used:

Retries

Players who retry after failure demonstrate perseverance.

Session Duration

Longer gameplay sessions indicate sustained engagement.

Pause Count

Frequent pauses may indicate loss of focus, hesitation, or disengagement.

Therefore pauses reduce persistence.

Interpretation

- More retries → Higher persistence
- Longer play time → Higher persistence
- More pauses → Lower persistence

Cognitive Skill Measured

- Motivation
- Resilience
- Sustained attention

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5. Consistency Rate

Formula

Combo Consistency = min(100, 100 × Max Combo / 50)

Stability = 100 − Error Rate

Pause Penalty = 20 × Pause Count / 5

Consistency Rate = max(0, min(100, 0.5 × Combo Consistency + 0.5 × Stability - Pause Penalty))

Why this Formula?

Consistency measures how stable and reliable performance remains throughout a session.

Two indicators are used:

Combo Consistency

A higher combo indicates the player can repeatedly perform successful actions without interruption.

Stability

A lower Error Rate means performance remains reliable over time.

Pause Penalty

Frequent pauses interrupt gameplay flow and indicate inconsistency.

Interpretation

- High combo streaks → Higher consistency

- Fewer mistakes → Higher consistency
- More pauses → Lower consistency

Cognitive Skill Measured

- Focus maintenance
- Self-regulation
- Performance stability

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6. Overall Performance Score

Formula

Overall Performance Score = 0.30 × Accuracy Rate + 0.25 × Response Rate + 0.15 × (100 − Error Rate) + 0.15 × Persistence Rate + 0.15 × Consistency Rate

Why this Formula?

The overall score combines all dimensions into a single performance indicator.

Weight Distribution

Accuracy Rate = 30%

Accuracy is the most important component because correct actions are fundamental to success.

Response Rate = 25%

Fast and effective reactions are central to Fruit Ninja gameplay.

Error Avoidance = 15%

Reducing mistakes improves overall quality of performance.

Persistence Rate = 15%

Shows willingness to continue and improve.

Consistency Rate = 15%

Rewards stable performance over time.

Interpretation

90 – 100 : Exceptional Performance

75 – 89 : Strong Performance

60 – 74 : Average Performance

40 – 59 : Needs Improvement

0 – 39 : Significant Improvement Needed

Assumptions

The following empirical benchmarks were derived from observed dataset limits:

- Maximum Session Duration = 300 seconds
- Maximum Combo = 50
- Maximum Retries = 3
- Response Speed Benchmark = 2.5 actions/second

Values exceeding benchmarks are capped at 100 to maintain normalization.

The raw Overall Score generated by the game engine was intentionally excluded from rate calculations to avoid circular dependencies and maintain interpretability. All derived metrics are calculated solely from observable gameplay behavior.